

Instructions, User Manuals, and Standard Operating Procedures

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COMMUNICATION AT WORK

This scenario shows the importance of effective instructions in the biomedical industry and various communication channels used to convey step-by-step procedures.

As the baby-boom generation ages, medical needs are expanding—sometimes faster than medical care facilities and medical professionals can manage. One area in which this has been felt most acutely is in medical laboratories. Thousands of medical technicians need to be trained to accommodate increased demand.

PhlebotomyDR is a medical consulting firm seeking to solve this problem. Its primary area of concern is training newly hired technicians responsible for performing blood collection. PhlebotomyDR facilitates training workshops to teach venipuncture standards and venipuncture procedures.

PhlebotomyDR focuses on the following venipuncture instructions:

- Proper patient identification procedures
- Proper equipment selection, sterilization, use, and cleaning
- Proper labeling procedures
- Order of phlebotomy draw

Learning Objectives

When you complete this chapter, you will be able to

- Organize your instruction effectively, including an introduction, table of contents, and discussion of sequenced steps
- Number your steps, and start each step with a verb
- Understand that hazard alerts are important to protect your reader and your company, so place these alerts early in your instruction or before the appropriate step
- Follow a chronological sequence in the instruction
- Know that screencasting, with a mixture of animated screen visuals, voiceovers, and captioned text, benefits your audience by allowing the audience to see the procedure in action
- Know that an SOP is a set of written instructions documenting routine or repetitive technical or administrative activities followed by business and industry to ensure accurate job performance, consistent quality, integrity of the end product, and adherence to governmental regulations
- Use highlighting techniques to emphasize important points, thus minimizing damage to equipment or injury to users
- Know that SOPs are used in many fields, including science, healthcare, biomedical technology, government, military, the computer industry, and others
- Consider usability testing to determine if your procedure is valid
- Evaluate instructions using the checklist

- Patient care before, during, and following venipuncture
- Safety and infection control procedures
- Procedures to follow when meeting quality assurance regulations

Each of the instructions mentioned requires numerous steps, complete with visual aids.

PhlebotomyDR offers its audience various communication channels. Hospitals, labs, and treatment centers can access PhlebotomyDR's instructions as follows:

- Hard-copy instructional manuals
- Online instructions, accessible at <http://www.phlebotomydr.com>
- Videos showing step-by-step performances of blood collection, complete with case studies enacted by technicians, patients, and supervisors
- Standard Operating Procedures used to achieve uniformity and efficiency

- Smartphone apps for individual tutorials
- One-on-one tutorials with trained phlebotomists
- Instructional workshops designed for groups of seminar participants

PhlebotomyDR's outstanding staff realizes that trained technicians make an enormous difference. Training, achieved through instructional manuals, electronic aids, and individual facilitation, ensures the health and safety of patients. Of equal importance, excellent training also benefits many stakeholders. Untrained technicians make errors that cost us all. Medical errors create insurance problems, the need to redo procedures, increased medical bills, the potential involvement of regulators and legislators, and dangerous repercussions for patients.

In contrast, effective communication, achieved through successful instruction, saves lives, time, and money.

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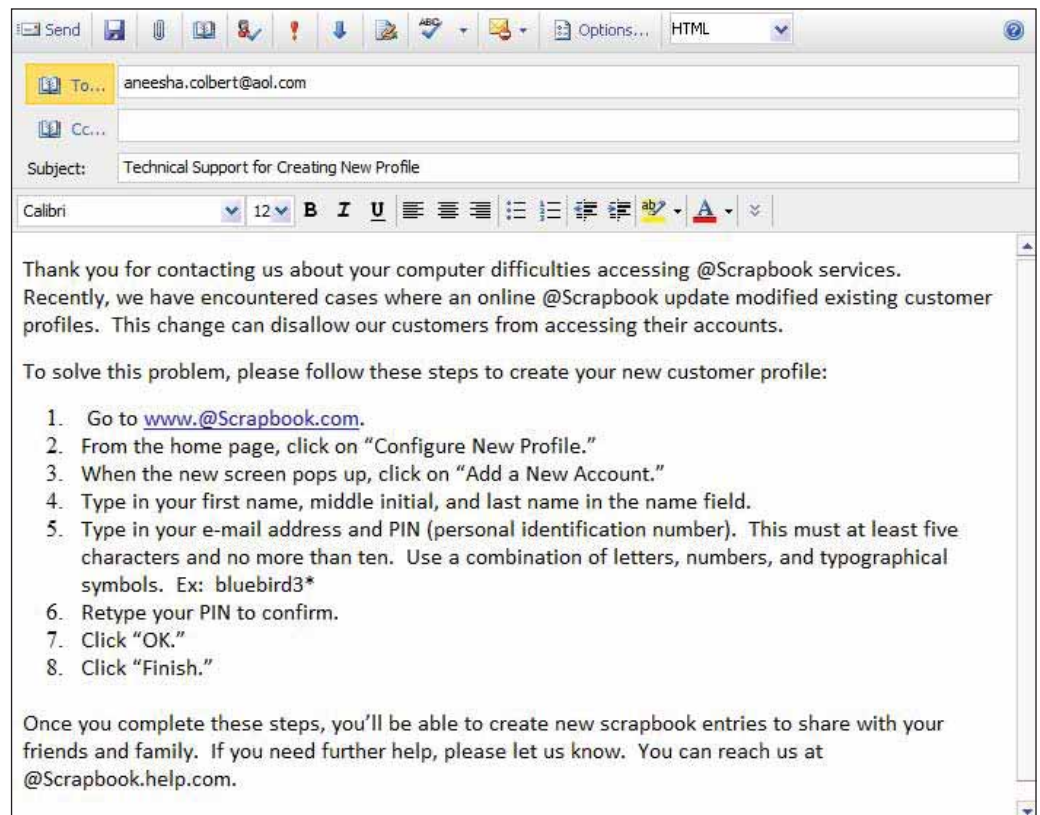
For excellent examples of instructions about home improvement, automobiles, gardening, crafts, and more, check out the following Do It Yourself Network link: <http://www.diynetwork.com/>.

Why Write an Instruction?

Almost every manufactured product comes complete with instructions. You will receive instructions for baking brownies; making pancakes; assembling children's toys; or changing a tire, the oil in your car, or the coolant in your engine. Instructions help people set up iPod and MP3 systems, construct electronic equipment, maintain computers, and operate fighter planes.

We frequently see short instructions about computer-related problems. If you are at home or in the office, for example, and you can't figure out how to operate one of your computer's applications, you can access online help or dial a 1-800 hotline to speak to a computer technician. Figure 1 is an e-mail message giving a customer the steps to follow for computer technical support.

FIGURE 1 Instructions for Computer Technical Support



Include instructions or user manuals whenever your audience needs to know how to

Operate a mechanism	Collect lab specimens
Install equipment	Use an autoclave
Manufacture a product	Service equipment
Package a product	Troubleshoot a system
Perform lab experiments	Use software
Test components	Set up a product
Maintain equipment	Implement a procedure
Clean a product	Assemble a product
Draw blood	Repair a system

Criteria for Writing Instructions

Odds are good that you've read a badly written instruction. You've probably encountered an instruction that failed to define unfamiliar terminology, didn't tell you what equipment you needed to perform the task, didn't warn you about dangers, didn't clarify how to perform certain steps, and didn't provide you enough graphics or a video to complete the job successfully. Usually, such poorly written instructions occur because writers fail to consider their audience's needs. Successful instructions, in contrast, start with audience recognition.

Audience Recognition

The instruction tells you to “place the belt on the motor pulley,” but you don't know what a “motor pulley” is nor do you know how to “place the belt” correctly. You are told to “discard the used liquid in a safe container,” but you don't know what is safe for this specific type of liquid. You are told to “size the cutting according to regular use,” but you have never regularly performed this activity before.

Here is a typical instruction written without considering the audience.

To overhaul the manual starter, proceed as follows: Remove the engine's top cover. Untie the starter rope at the anchor and allow the starter rope to slowly wind onto the pulley. Tie a knot on the end of the starter rope to prevent it from being pulled into the housing. Remove the pivot bolt and lift the manual starter assembly from the power head.

◀ EXAMPLE

Although many high-tech readers might be able to follow these instructions, many more readers will be confused. How do you remove the engine's top cover? Where is the anchor? Where is the pivot bolt, and how do you remove it? What is the power head?

The problem is caused by writers who assume that their readers have high-tech knowledge. This is a mistake for several reasons. First, even high-tech readers often need detailed information because technology changes daily. You cannot assume that every high-tech reader is up-to-date on these technical changes. Thus, you must clarify. Second, low-tech and lay readers—and that's most of us—carefully read each and every step, desperate for clear and thorough assistance.

As the writer, you should provide your readers with the clarity and thoroughness they require. To do this, recognize accurately who your readers are and give them what they want, whether that amounts to technical updates for high-tech readers or precise, even simple, information for low-tech or lay readers. The key to success as a writer of instructions is the following: Don't assume anything. Spell it all out—clearly and thoroughly.

Ethical Instructions

Your job as technical writer is to meet your audience's needs. This not only entails clarity in an instruction, but also it demands ethical behavior. User manuals and instructions are perfect examples of the importance of ethical communication.

Legalities in User Manuals. As a corporate employee, you never want your writing to lead to customer injuries or to cause a customer to damage equipment. Instead, to avoid issues of liability, your communication must identify the potential for harm. Clearly stated hazard notations, warranties, and disclaimers, discussed later in this chapter, allow you to warn your audience of potential dangers and to set limits and exceptions for product guarantees. By doing so, you adhere to your legal responsibilities to the company and to the client.

Practicalities in User Manuals. Ethical writing doesn't just keep customers safe. It also satisfies your customer's need for up-to-date information, and that's just good business. For example, let's say that a decade ago, your company wrote a technical specification for a product. The product has changed over the years, improving through numerous iterations. However, your company's customer support (training manuals, online help, user manuals, and troubleshooting guides) have not been revised to reflect these changes. Management says that the company doesn't have the time or budget to update the customer support. This can lead to many problems in the manuals, including inconsistencies, inaccuracies, irregularities, and customer complaints. If a client stops buying your products because the supporting documents are flawed, the company loses money. The practical solution, then, is to revise the manuals, bringing them up to date. Doing so is a company's ethical responsibility to its customers.

Components of Instructions

Not every instruction will contain the same components. Some very short instructions will consist of nothing more than a few, numbered steps. Other instructions, however, will consist of the components shown in Figure 2.

FIGURE 2 Key Components of Instructional Manuals

Title Page Topic Graphic Purpose	Hazards 	Table of Contents	Introduction	List of Required Tools/Equipment
Glossary of Terms	Steps 1. 2.	Steps 3. 4. Etc.	Additional Components	Corporate Contact Information

Title Page. Preface your instruction with a title page that consists of the *topic* about which you are writing, the *purpose* of the instruction, and a *graphic* depicting your product or service. For example, to merely title your instruction “iPod” would be uninformative. This title names the product, but it does not explain why the instruction is being written. Will the text discuss operating instructions, troubleshooting, service, or maintenance? A better title would be “Operating Instructions for the iPod.” Adding a graphical representation lets your audience see what the finished product will look like and helps you market your product or service.

Safety Requirements. You can place safety requirements anywhere throughout your text. If a particular step presents a danger to the reader, call attention to this hazard just before asking the reader to perform the step. Similarly, to help your audience complete an action, place a note before the step, suggesting the importance of using the correct tool, not overtightening a bolt, or wearing protective equipment.

In addition to placing safety requirements before a step, consider prefacing your entire instructions with hazard notations. By doing so, you make the audience aware of possible dangers, warnings, cautions, or notes in advance of performing the instructions. This is important to avoid potentially harming an individual or damaging equipment and to prevent costly lawsuits. Correctly using hazard notations will avoid expensive liability issues and adhere to ethical communication standards.

Access. Make the hazard notations obvious. To do so, vary your typeface and type size, use white space to separate the warning or caution from surrounding text, box the warning or caution, and call attention to the hazards through graphics.

Definitions. What does *caution* mean? How does it differ from *warning*, *danger*, or *note*?

Four primary organizations that seek to provide a standardized definition of terms are the American National Standards Institute (ANSI), the U.S. military (MILSPEC), the Occupational Safety and Health Administration (OSHA), and the MSDS (material safety data sheets) search national repository. To avoid confusion, we suggest the following hierarchy of definitions, which clarifies the degree of hazard:

1. **Note.** Important information, necessary to perform a task effectively or to avoid loss of data or inconvenience.
2. **Caution.** The potential for damage or destruction of equipment.
3. **Warning.** The potential for serious personal injury.
4. **Danger.** The potential for death.

Colors. Another way to emphasize your hazard message is through a colored window or text box around the word. Usually, *Note* is printed in blue or black, *Caution* in yellow, *Warning* in orange, and *Danger* in red.

Text. To further clarify your terminology, provide the readers text to accompany your hazard alert. Your text should have the following three parts:

1. **A one- or two-word identification alerting the reader.** Words such as “High Voltage,” “Hot Equipment,” “Sharp Objects,” or “Magnetic Parts,” for example, will warn your reader of potential dangers, warnings, or cautions.
2. **The consequences of the hazards, in three to five words.** Phrases like “Electrocution can kill,” “Can cause burns,” “Cuts can occur,” or “Can lead to data loss,” for example, will tell your readers the results stemming from the dangers, warnings, or cautions.
3. **Avoidance steps.** In three to five words, tell the readers how to avoid the consequences noted: “Wear rubber shoes,” “Don’t touch until cool,” “Wear protective gloves,” or “Keep disks away.”

Icons. Equipment is manufactured and sold globally; people speak different languages. Your hazard alert should contain an icon—a picture of the potential consequence—to help everyone understand the caution, warning, or danger.

Figure 3 shows an effective page layout and the necessary information to communicate hazard alerts.

Table of Contents. Your instruction might have several sections. In addition to the actual steps, the instructional manual could include technical specifications, warranties, guarantees, FAQs, troubleshooting tips, and customer service contact numbers. An effective table of contents will allow your readers to access any of these sections individually on an as-needed basis.

Introduction. Companies need customers. A user manual might be the only contact a company has with its customer. Therefore, instructions often are reader-friendly and seek to achieve audience recognition and audience involvement. The manuals try to reach customers in a personalized way. Look at the following introduction from a user manual:

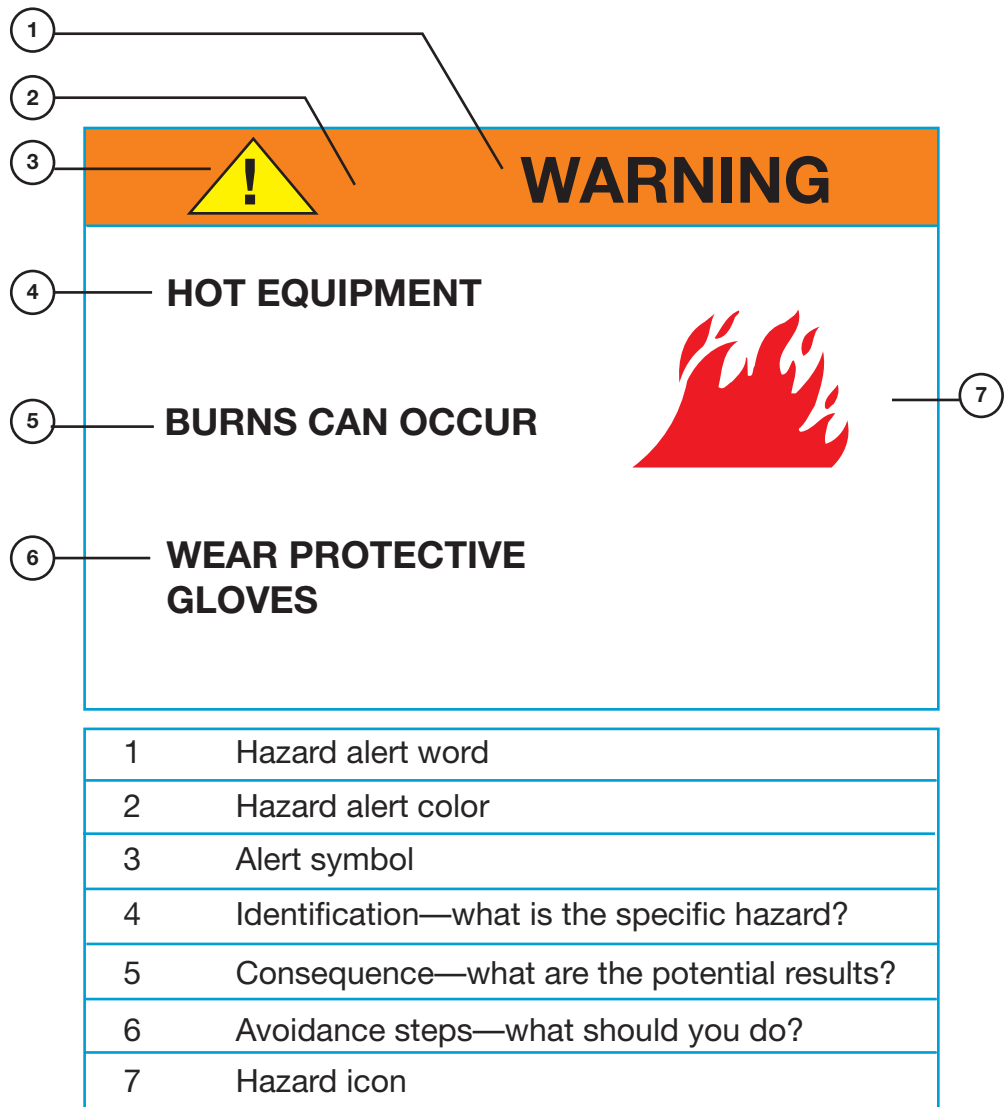
Thank you for your purchase. Installation and operating procedures are contained in this manual about your new 3-D TV. The minutes you spend reading these instructions will contribute to hours of viewing pleasure.

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For samples of hazard icons, check out the following link:

- http://www.speedysigns.com/signs/osha_readymade.asp

◀ EXAMPLE

FIGURE 3 Hazard Alert

This introduction uses pronouns (*you* and *your*) to personalize the manual. The introduction also uses positive words, such as *Thank you* and *pleasure* to achieve positive customer contact. An effective introduction promotes good customer-company relationships.

Glossary. If your instruction uses the abbreviations *BDC*, *CCW*, or *CPR*, will the readers know that you are referring to “bottom dead center,” “counterclockwise,” or “continuing property records”? If your audience is not familiar with your terminology, they might miss important information and perform an operation incorrectly.

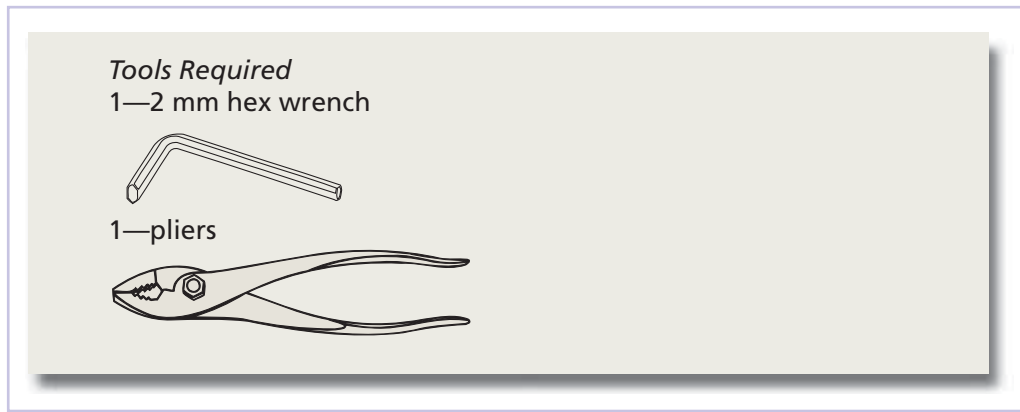
To avoid this problem, define your abbreviations, acronyms, or technical terms. You can define your terms early in the instruction, throughout the manual, or in a glossary located at the end of your manual. The following example defines terms alphabetically in a glossary:

GLOSSARY

BDC	Bottom dead center
CCW	Counterclockwise
Danger	This hazard alert designates the possibility of death. Be extremely careful when performing an operation.
RMS	Root mean square

◀ **EXAMPLE**

Required Tools or Equipment. What tools or equipment will the audience need to perform the procedures? You don't want your audience to be in the middle of performing a step and suddenly realize that they need a missing piece of equipment. Provide this important information either through a list or graphics depicting the tools or equipment necessary to complete the tasks.

◀ **EXAMPLE**

Instructional Steps. The instructional steps are the most important part of your manual—the actual actions required of the audience to complete a task. To successfully write your instruction, follow these steps.

Organize the Steps Chronologically. You cannot tell your readers to do step 6, then go back to step 2, then accomplish step 12, and then do step 4. Such a distorted sequence would fail to accomplish the task. To operate machinery, monitor a system, troubleshoot equipment, or perform a standard operating procedure, your readers must follow a chronological sequence. Be sure that your instruction is chronologically accurate.

Number Your Steps. Do not use bullets or the alphabet. Numbers, which you can never run out of, help your readers refer to the correct step. In contrast, if you used bullets, your readers would have to count to locate steps—seven bullets for step 7, and so on. If you used the alphabet, you'd be in trouble when you reached step 27.

Use Highlighting Techniques. You can use boldface, different font sizes and styles, text boxes, emphatic warning words, color, or italics to call attention to special concerns. A danger, caution, warning, or specially required technique must be evident to your reader. If this special concern is buried in a block of unappealing text, it will not be read. This could be dangerous to your reader or costly to you and your company. To avoid lawsuits or to help your readers see what is important, call it out through formatting.

Limit the Information within Each Step. Don't overload your reader by writing lengthy steps.

BEFORE**Overloaded Steps**

Start the engine and run it to idling speed while opening the radiator cap and inserting the measuring gauge until the red ball within the glass tube floats either to the acceptable green range or to the dangerous red line.

AFTER**Separated Steps**

1. Start the engine.
2. Run the engine to idling speed.
3. Open the radiator cap.
4. Insert the measuring gauge.
5. Determine whether the red ball within the glass tube floats to the acceptable green range or up to the dangerous red line.

Develop Your Points Thoroughly. Avoid vague content by clarifying directions, needed equipment, and cautions.

The After example clarifies what is meant by correct rotating and approved lubricant. The steps are also separated for enhanced readability.

BEFORE

After rotating the discs correctly, grease each with an approved lubricant.

AFTER

1. Rotate the disks clockwise so that the tabs on the outside edges align.
2. Lubricate the discs with 2 oz of XYZ grease.

Use Short Words, Sentences, and Paragraphs. Help your audience complete the task quickly and easily. People read instructions because they need help. It's hard to complete a task when they are unfamiliar with equipment, tools, dangers, and technical concepts. You don't want to compound their challenge with long words, sentences, and paragraphs.

Begin Your Steps with Verbs. Note that each of the numbered steps in the following example begins with a verb.

EXAMPLE ►**Verbs Begin Steps**

1. *Organize* the steps chronologically.
2. *Number* the steps.
3. *Use* highlighting techniques.
4. *Limit* the information within each step.
5. *Develop* the points thoroughly.
6. *Use* short words and phrases.
7. *Begin* the steps with verbs

Do Not Omit Articles. Articles, such as *a*, *an*, and *the*, are part of standard English. Although you might see instructions that omit these articles, please don't do so yourself. Articles do not take up much room in your text, but they make your sentences grammatically correct.

BEFORE

1. Press right arrow button to scroll through list of programs.
2. Select program to scan.
3. Place item to scan face down on scanner glass in upper left corner.

AFTER

1. Press the right arrow button to scroll through a list of programs.
2. Select the program you want to scan.
3. Place an item to scan face down on the scanner glass in the upper left corner.

Additional Components. Your instruction might include the following additional components.

Technical descriptions. In addition to the step-by-step instructions, many manuals contain technical descriptions of the product or system. A description could be a part-by-part

explanation or labeling of a product or system's components. Such a description helps readers recognize parts when they are referred to in the instruction. For example, if the user manual tells the reader to lay shingles with the tabs pointing up, but the reader doesn't know what a "tab" is, the step cannot be performed. In contrast, if a description with appropriate callouts is provided, then the reader's job has been simplified.

Perhaps the user manual will contain a list of the product's specifications, such as size, shape, capacity, capability, and materials of construction. Specifications allow the user to decide whether the product meets the reader's needs. Does it have the desired resolution? Is it the preferred weight and size? The specification for a smartphone in the following example answers these questions.

◀ EXAMPLE

Smartphone Specifications

Height:	114 mm (4.48 inches)
Width:	66 mm (2.6 inches)
Depth:	15 mm (0.59 inches)
Weight:	136 grams (4.8 ounces)
Monitor Resolution:	480 × 320 pixel color display

Finally, a user manual might include a schematic, depicting the product or system's electrical layout. This would help the readers troubleshoot the mechanism.

Warranties. Warranties protect the customer and the manufacturer. Many warranties tell the customer, "This warranty gives you specific legal rights, and you may also have other rights that vary from state to state." A warranty protects the customer if a product malfunctions sooner than the manufacturer suggests it might: "This warrants your product against defects due to faulty material or installation." In such a case, the customer usually has a right to free repairs or a replacement of the product.

However, the warranty also protects the manufacturer. No product lasts forever, under all conditions. If the product malfunctions after a period of time designated by the manufacturer, then the customer is responsible for the cost of repairs or a replacement product. The designated period of time differs from product to product. Furthermore, many warranties tell the customers, "This warranty does not include damage to the product resulting from normal wear, accident, or misuse."

Disclaimers are another common part of warranties that protect the manufacturer. For example, some warranties include the following disclaimers:

- Note: Any changes or modifications to this system not expressly approved in this manual could void your warranty.
- Proof of purchase with a receipt clearly noting that this unit is under warranty must be presented.
- The warranty is only valid if the serial number appears on the product.
- The manufacturer of this product will not be liable for damages caused by failure to comply with the installation instructions enclosed within this manual.

Accessories. A company always tries to increase its income. One way to do so is by selling the customer additional equipment. A user manual promotes such equipment in an accessories list. This equipment isn't mandatory, required for the product's operation. Instead, an accessories list offers customers equipment such as extra-long cables or cords, carrying cases, long-life rechargeable batteries, belt clips and cases, wireless headsets, automotive chargers, and memory cards. Often the specifications are also provided for these accessories.

Frequently asked questions. Why take up your customer support employees' valuable time by having them answer the same questions over and over? By including a frequently asked

questions (FAQs) page in the user manual, common consumer concerns can be addressed immediately. This will save your company time and money while improving customer relations.

Corporate contact information. Conclude your manual by providing your company's street, city, and state address; 1-800 hotline telephone number; Web site URL; e-mail address; fax number; social media sites; or other ways your audience could contact the company with questions or requests for more information.

Graphics. Clarify your points graphically. Use drawings, photographs, and screen captures that are big, simple, clear, keyed to the text, and labeled accurately. Not only do these graphics make your instructions more visually appealing, but also they help your readers and you. What the reader has difficulty understanding, or you have difficulty writing clearly, your graphic can help explain pictorially.

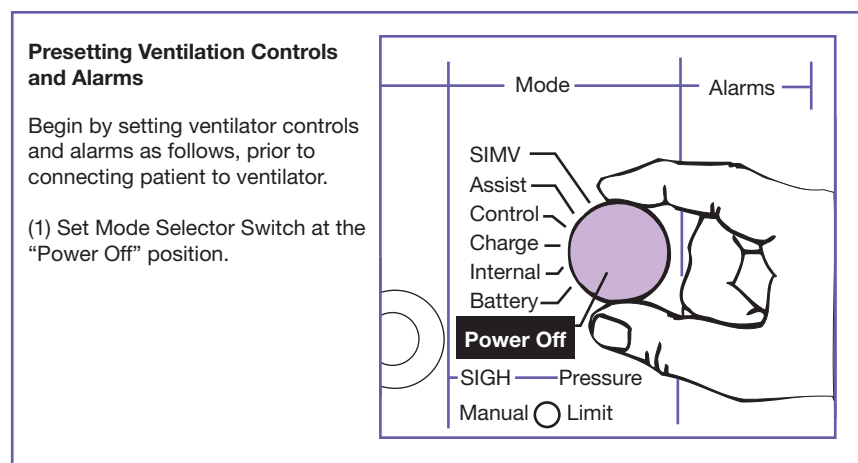
A company's use of graphics often depends on the audience. With a high-tech reader, the graphic might not be needed. However, with a low-tech reader, the graphic is used to help clarify. Look at the examples of instructions for the same procedure. Table 1 uses text without graphics for the high-tech audience. Figure 4 uses text with graphics to help the low-tech audience better understand the required steps.

TABLE 1 Text without Graphics for High-Tech Readers

Location Item	Action	Remarks
Note: For additional operating instructions, refer to Companion 2800 operator's manual.		
1. MODE switch	Set to POWER OFF.	
2. Rear panel	Ensure that instructions on all labels are observed.	
3. Patient tubing circuit	Assemble and connect to the unit.	See Table 2, step 3.
4. Power cord	Connect to a 120/220 V AC grounded outlet.	If the power source is an external battery, connect the external battery cable to the unit per Table 2, step 4.
5. Cascade I humidifier	Connect to the unit.	See Table 2, step 2.

Source: Courtesy of Puritan Bennett Corp.

FIGURE 4 Instructions with Large, Bold Graphics for Low-Tech Audience



TECHNOLOGY TIPS

Using Screen Captures to Visually Depict Steps

Here's a good, simple way to visually depict steps in your instruction—screen captures. A screen capture lets you copy any image present on your computer monitor. To make screen captures, follow these steps.

Capturing the Image

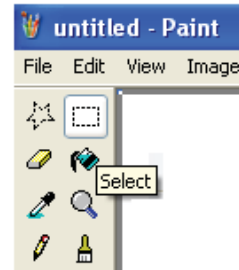
1. Find the graphic you want to include in your instructions.
2. Press the **Print Screen** key on your computer keyboard.



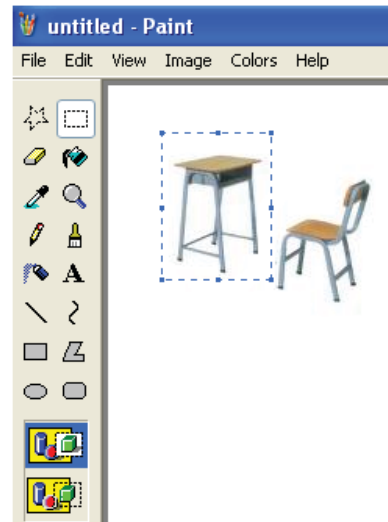
Cropping the Image

Pressing the print screen key captures the entire image seen on the monitor. This might include images that you do not want to include in your instruction. To crop your image, follow these steps:

1. Paste the **Print Screen** image into a graphics program like Microsoft Paint.
2. Click the **Select** icon.



3. Click and drag the **Select** tool to choose the part of the captured graphic you want to crop.



4. Copy and paste this cropped image into your user manual.



"Selected" portion of the graphic chosen for cropping

Note: When capturing screen shots, be careful to avoid infringing upon a company's copyright to the image. Make sure that your use of the image meets the principle of "fair use" (which permits the use of images for criticism, comment, news reporting, teaching, scholarship, or research—Copyright Act of 1976, 17 U.S.C. § 107).

Collaboration to Create User Manuals

Collaboration is essential for team members who write user manuals. User manuals are often long and complex, requiring input from a variety of individuals, each with a different job function. Stacy Gerson, technical writer for GBA Master Series (gbaMS), works collaboratively with multiple team members when writing her user manuals. Her team consists of three separate cohorts: other writers, clients, and a project team of subject matter experts.

The subject matter experts are headed by a project manager. This person is in charge of the budget and schedule for the user manual creation. The manager oversees the following:

- Account specialists, who liaison with the clients to determine customer needs
- Business analysts, who determine what software changes need to be made to meet the clients' expectations
- Software developers, who code the software programs
- Quality assurance personnel, who make sure the coding is correct

Stacy takes the information from the above team members and writes the manual. If the manual is very long, she might work with a team of writers, most of whom are dispersed in various cities. Team members collaborate virtually through online chats, text messages, e-mail, and telephone calls. They discuss how the manual will look, mock up parts of the project, decide how to delegate responsibilities, work on their individual parts, proofread each other's work, and then merge the components for the finished project.

Finally, Stacy meets with a representative group of clients who complete the steps within the manual for usability testing. By gauging the challenges her test group might encounter, Stacy can correct glitches and fine tune the manual. Once this is completed, the manual is ready for publication, either in hard-copy or electronic form.

Instructional Videos

Another way to depict your instructional steps graphically is through videos. Screencasting, a mixture of animated screen visuals, voiceovers, and captioned text, will benefit your audience in the following ways:

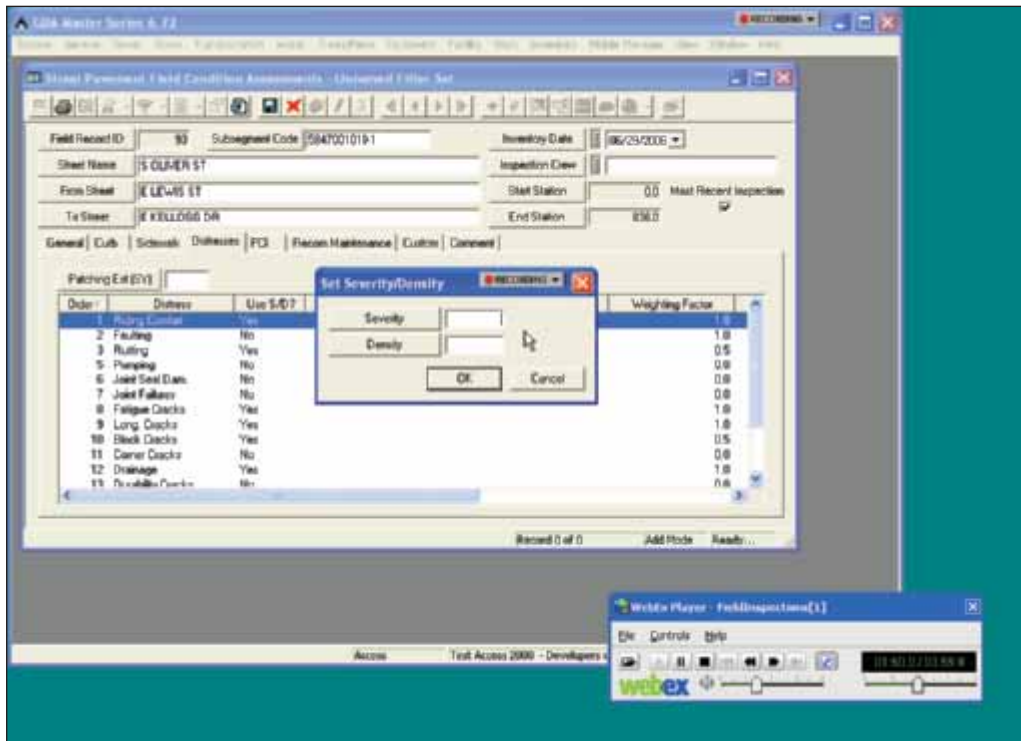
- Animated videos show the end-user how to perform a step more successfully than static text might.
- Videos are an excellent communication channel for visual learners.
- Through podcasting and smartphone apps, instructional videos can be downloaded anywhere, anytime by mobile users.
- A company can enhance its marketing strategy by advertising its instructional videos as an alternative communication channel to hard-copy text. YouTube is an excellent social networking tool for sharing instructional videos and for marketing a company's product.
- Instructional videos allow for end-user interaction. The audience can fast forward, slow down, go backward, or freeze screens to learn how to perform tasks at their own pace.
- A video can be more informative than static text and photography. A video screencast shows how an end-user interacts with machinery, equipment, tools, and more.
- Video instructions help end-users see the consequences of hazard alerts. For example, the user can see a short animation showing the results of pushing the wrong button, using the incorrect equipment, or putting the wrong chemical in a test tube.

- You can make instructional videos using a variety of tools that are supported by Windows, Mac, or Linux. These include WebEx, Camtasia, Captivate, CamStudio, iShowU, SnapzPro, Wink, and Encorder.

(Tietjen; Sharp)

Figure 5 shows a screen capture of one step in a video instruction.

FIGURE 5 Video Instruction with Voiceover Narration



Voiceover Narration:

The first step in assessing street pavement field conditions is to determine the level of riding comfort. Click on **Riding Comfort** to select this option. The **Set Severity/Density** box drops down. Now, fill in the **Severity** and **Density** fields and click **OK**. This takes you to the next step.

At the bottom of the video instruction, GBA Master Series provides a WebEx Player control box. With these buttons and slides, the end-user can control the sound, fast forward, rewind, or pause the video. The video counter allows the end-user to skip ahead or return to other portions of the video.

SPOTLIGHT

How GBA Master Series Uses Screencasting for Instructions

Technical writer Stacy Gerson writes online user manuals and training guides in hard-copy and online video formats. Her audience includes office workers, dispatchers, maintenance supervisors, field workers, engineers, and employees of citizen complaint call centers.

These manuals and videos help her clients use gbaMS's software related to public works preventative maintenance scheduling, work orders, asset inventory, asset inspections, employee timesheets, and administrative modules for city, county, and state planning and budgeting.

To write her user manuals and online videos, Stacy

- Interacts with her programmers, asking questions to clarify challenging content.
- Watches WebEx movies created by the software developers. These movies walk through programs and



provide voiceover explanations of the content. Stacy uses the movies for the basis of her text for both the hard-copy manuals and video scripts.

- Digs through the complex coding, provided by the subject matter experts to find what will be most useful for her lay readers.
- "Learns how to use the software by using it."

Stacy focuses on the big picture when she creates her online manuals and videos. Her goals are to create easy-to-read text, easy-to-follow steps, and material that "is not just well written but content that will help the end-user." As she adapts her scripts to online video instruction, she adds screen shots and provides a voiceover narration to explain which fields to fill in on the screens and what step to perform next.

Stacy has found that she needs to write well, have grammatical abilities, and empathize with the end-users' needs. She also has to consider whether or not her online videos use enough narration to explain the screen shots.

Her number one task is to make things easily understandable for gbaMS's customers. The software can be complex at times, so if

an instruction is unclear, Stacy says, "the customer can't do the job." To make the instructions understandable, Stacy explains, "I try to write text for the manuals and the videos that is logical, clear, simple, and visually appealing. If I look at something and say, 'What's that?' then I have to assume that it won't be clear to the client either."

Standard Operating Procedures

Standard Operating Procedures (SOPs) are a common form of instruction in many scientific and technological fields.

Why Write an SOP?

An SOP is a set of written instructions that documents routine or repetitive technical or administrative activities followed by business and industry. SOPs ensure accurate job performance and consistent quality and integrity of the end product. SOPs also ensure that governmental regulations are followed. SOPs must be reviewed and enforced by management. Employees need to reference SOPs for accuracy of procedures, so SOPs must be available as hard-copy documents and in electronic formats. Annual reviews of SOPs by both management and employees are necessary to ensure that they are being followed.

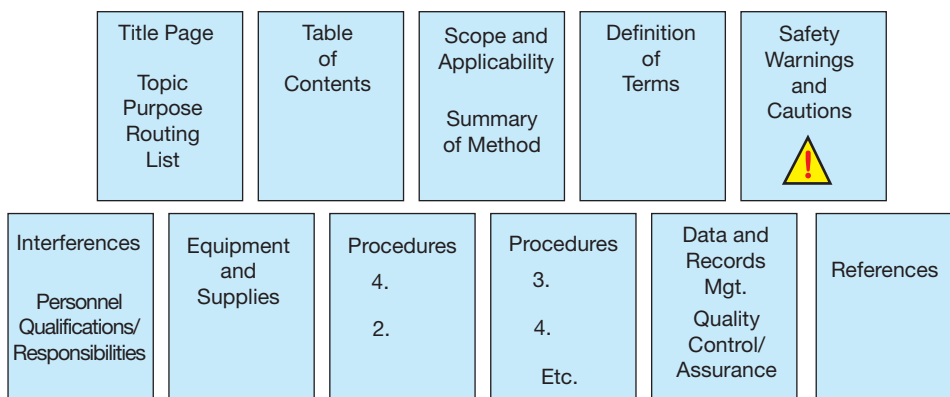
SOPs are used in many fields, including science, healthcare, biomedical technology, government, military, the computer industry, and others. These industries use SOPs to document how to do the following:

- Calibrate and standardize instruments
- Collect lab samples
- Handle and preserve food
- Analyze test data
- Troubleshoot equipment, machinery, and procedures
- List mathematical steps to follow for acquiring data and making calculations
- Assess hardware and software analytical data

Components of SOPs

SOPs contain many of the same components as user manuals already discussed in this chapter, such as procedural steps. However, several of the parts of an SOP are different. Figure 6 shows the key components of an SOP.

FIGURE 6 Key Components of SOPs



Title Page. Preface your SOP with a title page that consists of the topic about which you are writing, the date of the SOP, the purpose of the SOP, and the routing list of people who have to sign off on this procedure.

◀ **EXAMPLE**

Standard Operating Procedure

January 14, 2014

Preparing and Processing Algae Samples to Ensure Proper pH

Author: _____

Date Approved: _____

Manager: _____

Date Approved: _____

Quality Assurance Manager: _____

Date Approved: _____

Scope and Applicability. This section of the SOP can be limited to approximately three to five sentences. However, some SOPs, due to the complexity of the subject matter, can have Scope and Applicability sections that are several paragraphs long. This section of the SOP provides an overview for the audience. To do so, explain why the SOP has been written, show how it meets regulatory requirements, specify any limits for the use of the procedure, and state the applicability of the procedure.

◀ **EXAMPLE**

Scope

The purpose of this SOP is to establish uniform procedures for water compliance inspections (WCIs) performed by the Georgia Science and Assessment Division (GSAD). The WCIs evaluate the effectiveness and reliability of the state agency's inspection procedures to meet regulations of the Clean Water Act (CWA). The SOP will ensure thoroughness of all water compliance inspections and reports. The inspector may alter SOPs due to unexpected or unique problems in the field. Deviations from the SOP must be reported.

Applicability

The policies and procedures of the SOP apply to all personnel who take part in the WCIs.

Summary of Method. In this brief summary of the procedure (three to five sentences or short paragraphs), you might focus on any of the following, determined by the topic of the SOP:

- Title of people involved
- Their roles and duties
- Sequence of their involvement
- Sequence of activities performed

◀ **EXAMPLE**

Summary of the Method

During the WCI, the state inspector is the lead person supervising all tests. The state inspector manages a staff of scientists who gather water samples either manually or automatically with a portable compositor. Once the samples are collected, the state scientists divide the collection into containers, preserve the samples in ice, and ship the collection to a state laboratory for analysis. The laboratory technicians compare

the test results to acceptable limits to determine compliance. The state inspector must sign off on all parts of the procedure.

Once these tests are completed, the WCI inspector and the United States EPA inspector discuss procedures, findings, variances, and corrective actions. The EPA inspector fills out the Water Compliance Overview Inspection Checklist. Both inspectors are required to sign this form, which is kept on file in the state and national archives.

Interferences. Interferences include any components of the process that may interfere with the accuracy of the final product. This can include but is not limited to the following:

- Humidity
- Temperature
- Depth
- Altitude
- Weather
- Cleanliness
- Carelessness
- Sample size
- Contamination

EXAMPLE ►

Interferences

If the water has high concentrations of chlorine, this can interfere with test results. Therefore, if the inspector determines that chlorine is present, add sodium sulfide to the sample bottle before autoclaving.

Personnel Qualifications/Responsibilities. This section lists the required experience and certifications of the individuals performing the SOP. These qualifications can include education, years of experience, courses taken, continuing education hours completed, and more.

EXAMPLE ►

Personnel Qualifications/Responsibilities

This SOP is written specifically for water compliance inspections (WCIs) performed by the Georgia Science and Assessment Division (GSAD). All personnel who perform this activity must have the Basic Inspector Training for Water Compliance certification and have taken the state-required, 8-hour continuing education class. Additional training qualifications include GSAD regulation certification as well as on-the-job training.

Personnel performing this SOP have the following responsibilities:

1. WCI inspector—this person must have a minimum of a Bachelor of Science in chemistry (or related discipline), five years field work experience, three years laboratory work, and must be certified in state water compliance regulations. This inspector is responsible for coordinating staff and conducting the WCI test to meet deadlines. The inspector must be able to enforce the policies and procedures outlined in this SOP.
2. Scientists and laboratory technicians—these employees must have a minimum of an Associate of Science in chemistry, biotechnology, or a related field. These employees also must have passed state-level training in water compliance regulations. Scientists and laboratory technicians must meet deadlines for the WCI, submit all samples in accordance with regulatory policies, and adhere to the SOP.

Equipment and Supplies. This section can include the following items necessary to perform the SOP:

- Tools
- Equipment
- Reagents needed for the procedure
- Standards for the chemicals
- Biological specimens

If the steps in the process break down into multiple sections, you can divide the equipment and supplies into the units in a chronological pattern of organization. Remember to include quantities of items if needed to perform different tests in the process.

Equipment and Supplies

1.1 Sample Collection

- 1.1.1 Sterile sample jars
- 1.1.2 Syringes with carbon filters
- 1.1.3 Sterile screw-top lids

1.2 Lab Analysis

- 1.2.1 Air incubator
- 1.2.2 Water bath
- 1.2.3 Disposable sterile pipettes
- 1.2.4 Sterile applicator sticks
- 1.2.5 Sterile culture tubes
- 1.2.6 Sterile screw-cap tops
- 1.2.7 Pyrex graduated cylinders
- 1.2.8 Culture tube racks

◀ EXAMPLE

Data and Records Management. Include the following information in this unit of the SOP:

- Calculations to be performed during the procedure
- Forms for the reports
- Required reports
- Reporting intervals
- Report recipients
- Process to follow for recording and storing data and information generated by the SOP

Records Management

1. Appropriate SOPs will be kept in the laboratory library in black binders. All SOPs will be available to management and employees. When an SOP is removed from the library, sign your name and add the checkout date on the posted SOP inventory sheet.
2. No SOPs can leave the laboratory area.
3. The laboratory director will update the SOP binders as procedures or regulations change.
4. Staff will read all revised SOPs within five working days and sign and date the updated SOP list.

◀ EXAMPLE

Quality Control and Quality Assurance. Quality control activities help you ensure that you are checking for the highest quality and consistency of the procedure.

- Explain how you will verify your work through quality control. What self-checks will you implement? Will you retest, recount, or recalibrate findings?
- Include the specific process to follow for retesting the procedure, recounting to determine the performance level of the procedure, or recalibrating the machinery.
- Include the self-check intervals for retesting, recounting, or recalibrating. Detail the procedure for dealing with divergent results found through the self-checks.
- State how and to whom you will report the results of the quality control activities.

EXAMPLE ►

Quality Control

To achieve quality assurance, the laboratory will follow this procedure:

- **Sterility**—each sample is incubated for 24 hours at 0.5°C and examined for contaminants. If a sample tests positive for contaminants, the entire lot is discarded.
- **pH**—if the pH does not meet manufacturer’s specifications prior to autoclaving, pH can be adjusted according to EPA specifications. If the pH does not meet specifications after autoclaving, the entire sample must be discarded.
- **Testing**—the laboratory will test samples twice a year (January and June).

References. In this unit, cite the documents or procedures used in or referred to in your SOP. You should include documentation of additional SOPs referenced, literature used for research, and any additional instruction manuals used for reference. Attach copies of the documents if they are not readily available to your potential audience.

Test for Usability

You’ve written the instruction or SOP, but is it usable? If the instruction doesn’t help your audience complete the task, what have you accomplished? To determine the success of your manual, test the usability of the instruction or SOP as follows:

1. **Select a test audience.** The best audience for usability testing would include a representative sampling of individuals with differing levels of expertise. A review team composed only of high-tech readers would skew your findings. In contrast, a review team consisting of high-tech, low-tech, and lay readers would give you more reliable feedback.
2. **Ask the audience to test the instructions.** The audience members would attempt to complete the instructions, following the procedure step by step.
3. **Monitor the audience.** What challenges do the instructions seem to present? For example, has the audience completed all steps easily? Are the correct tools and equipment listed? Are terms defined as needed and presented where they are necessary? Has the test audience abided by the hazard notations? Are any steps overloaded and, thus, too complicated? Has each step been explained specifically?
4. **Time the team members.** How long does it take each member to complete the procedure? More important, why has it taken some team members longer to complete the task?
5. **Quantify the audience’s responses.** Once your test audience has completed the procedure, debrief these individuals to determine what problems they encountered.

Use the Instruction, User Manual, and SOP Usability Checklist to help gather quantifiable information about the instruction’s usability.

INSTRUCTION, USER MANUAL, AND SOP USABILITY CHECKLIST

Audience Recognition

- _____ 1. Are technical terms defined?
- _____ 2. Are graphics used to explain difficult steps at the reader's level of understanding?
- _____ 3. Do the graphics depict correct completion of difficult steps at the reader's level of understanding?
- _____ 4. Are the tone and word usage appropriate for the intended audience?
- _____ 5. Does the introduction involve the audience, clarify how the reader will benefit, or provide the scope and applicability of the procedure?

Development

- _____ 1. Are steps precisely developed?
- _____ 2. Is all required information provided, including hazards, technical descriptions, warranties, accessories, and required equipment or tools?

Conciseness

- _____ 1. Are words, sentences, and paragraphs concise and to the point?
- _____ 2. Does each step specify one action versus overloading the reader?

Consistency

- _____ 1. Is a consistent and parallel hierarchy of headings used?
- _____ 2. Are graphics presented consistently (same location, same use of figure titles and numbers, similar sizes, etc.)?

- _____ 3. Does wording mean the same throughout (technical terms, cautions, warnings, notes, etc.)?
- _____ 4. Is the same system of numbering used throughout?

Ease of Use

- _____ 1. Can readers easily find what they want through the following?
 - Table of contents
 - Glossary
 - Hierarchical headings
 - Headers and footers
 - Index
 - Cross-referencing or hypertext links
 - Frequently asked questions (FAQs)

Document Design

- _____ 1. Do graphics depict how to perform steps?
- _____ 2. Is white space used to make information accessible?
- _____ 3. Does color emphasize hazards, key terms, or important parts of a step?
- _____ 4. Do numbers or bullets divide steps into manageable chunks?
- _____ 5. Are hazards (dangers, warnings, cautions, and notes) boldfaced, typed in appropriate colors, and/or placed in textboxes to emphasize importance?

Sample Instructions

See Figure 7, Hewlett Packard OfficeJet Pro, for a sample instruction.

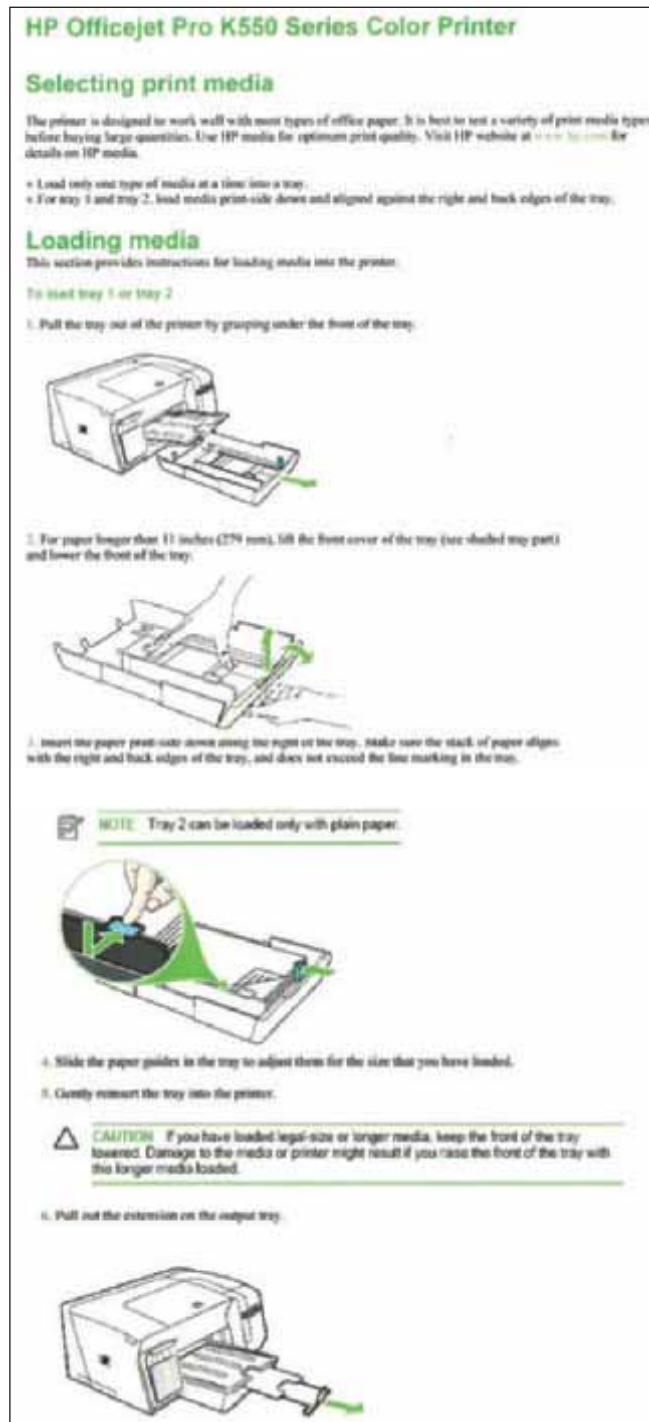
The Writing Process at Work

Now that you know what should be included in an instruction or SOP, it is time to write. The best way to accomplish your technical communication challenge is by following a process: Prewrite to gather data and determine objectives, write a rough draft of your instruction, and rewrite to revise.

Following is an example of an instruction written by Stacy Gerson, technical writer at gbaMS. To write this instruction, Stacy gathered data, determined objectives and sequenced the instruction through prewriting, drafted an instruction in writing, and finally revised the draft.

Prewriting

Flowcharts chronologically trace the stages of an instruction, visually revealing the flow of action, decision, authority, responsibility, input/output, preparation, and termination of process. Flowcharting is not just a graphic way to help you gather data and sequence your instruction, however. It is two-dimensional writing. It provides your reader content as well

FIGURE 7 HP Officejet Pro K550 Series Color Printer

Source: Courtesy of Hewlett-Packard

as a panoramic view of an entire sequence. Rather than just reading an instruction step by step, having little idea what will happen next, with a flowchart your reader can figuratively stand above the instruction and see where it's going. With a flowchart, readers can anticipate cautions, dangers, and warnings, since they can see at a glance where the steps lead.

To create a flowchart, use the International Organization for Standardization (ISO) flowchart symbols shown in Figure 8.

Figure 9 illustrates Stacy's flowcharting to gather data and to arrange it chronologically.

FIGURE 8 Flowcharting Symbols

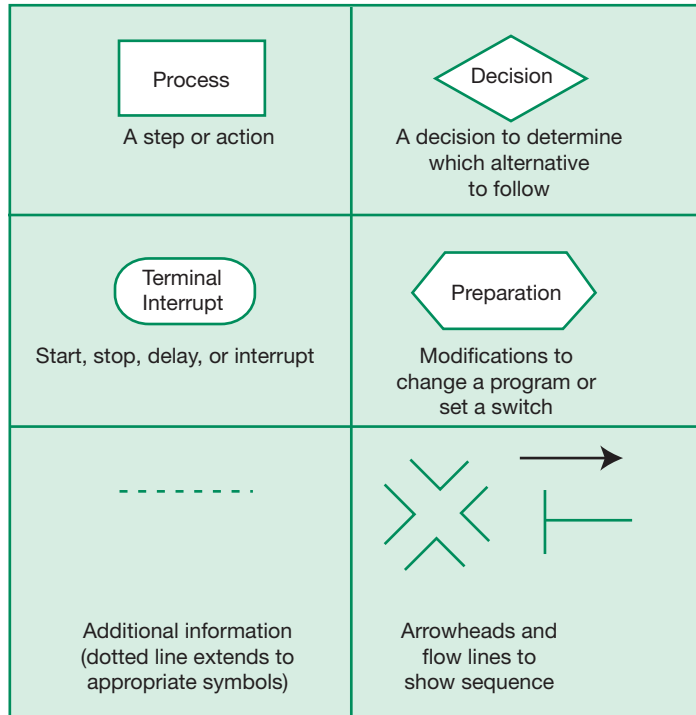
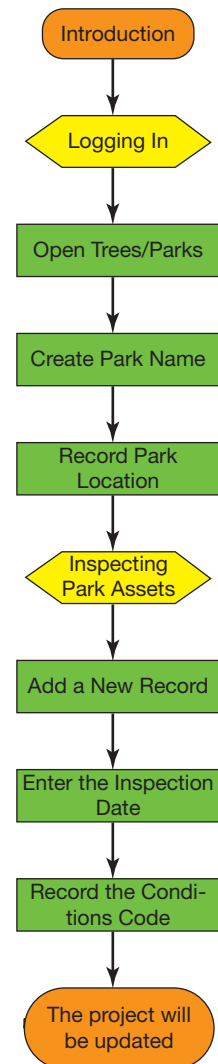


FIGURE 9 Flowchart for gbaMS Parks Inventory



Writing

Once you've graphically depicted your instruction sequence using a flowchart, the next step is to write a rough draft of the instruction. After completing the prewriting, Stacy composed a rough draft. A teammate reviewed her draft and made revision suggestions (Figure 10).

Rewriting

Rewriting, the last stage in the process, is the time to revise your instruction. Stacy revised the rough draft. The revision is shown in Figure 11.

FIGURE 10 gbaMS Instruction Rough Draft with Peer Reviewer Commentary

Add a title page, a table of contents for reader ease of access, a glossary to define terms the audience might not know, and a welcoming introduction.

The first step in the rough draft is actually three, separate steps. The use of the bullet and the numbering confused the test audience, so revise that setup.

You added unnecessary transitional words, such as "then," "next," and "finally." Remove these words since the numbered steps act as transition.

Add screen captures to help the audience understand the procedure.

Park Inventory

The *Parks* module is the most important module in the *Park Master*. All other park asset modules are dependent upon the *Parks* module. While you can access the other asset modules through the gbaMS main menu, it's best to add park assets directly from the *Parks* module. This allows the system to carry over linking information and details about the park. In the following step-by-step example, we'll show you how to create a park, record its location and attributes, and associate park assets.

1. Open **Trees/Parks>>Park Master>>Parks** from the gbaMS main menu.
 - Click **GO** to bypass the filter. When the module is opened, click the **Add** button in the toolbar to access a blank record.
2. Then, in the module header, create a unique Park Name and Alternate Park ID. Make sure to choose a name and ID that are recognizable; you'll need to be able to select the appropriate park name from a pick list in the park asset modules.
3. Next, record the park location, park supervisor, park type, district, region, owner, acquisition date, and opening date on the General tab.
4. Finally, select a Default WO Category. The category you select here will be carried over to any work order created from this record.

Inspecting Park Assets

All of our park asset inventory modules have associated inspection modules. In the following example, we'll show you how to add an inspection record for your park asset.

1. On the Inspections/Associations tab of the *Park Playground Equipment Inventory* module, right click in the Inspections grid and select **Add Record**.
2. The *Park Playground Equipment Inspection* module will open. There, you'll see the Equipment Number, Alternate Equipment ID, and Equipment Name carried over from the related inventory record.
3. Next, you'll need to enter the inspection date, inspection crew, and next scheduled inspection date.
4. Two Condition tabs are provided. You'll use these to record the condition codes (excellent, good, fair, and poor) for the various aspects of the playground equipment. These include the overall condition, stability, concrete exposure, protective surface, clearance, supports, and fastenings, among other things.

FIGURE 11 gbaMS Parks Inventory Completed Instruction



Source: Courtesy of gbaMS

TABLE OF CONTENTS

Introduction	1
Required Hardware and Software	2
Glossary	2
Park Inventory	3
Inspecting Park Assets	4

LIST OF ILLUSTRATIONS

FIGURE 1 Name and ID	3
FIGURE 2 Add Record	4
FIGURE 3 Equipment Name and Number	4
FIGURE 4 Equipment Condition	5

INTRODUCTION

In this workbook, we'll introduce you to the modules in the *GBA Parks Master™*. Using a series of step-by-step examples, we'll show you how to track and maintain your park assets. These include

- Parking lots
- Paths
- Fields and courts
- Landscaping
- Playground equipment
- Pools
- Furniture (and more)

Our software modules will help you manage your parks effectively and efficiently. While the *Parks Master's* multiple inventory and inspection modules track unique park assets, the steps provided in this workbook will apply equally to all park asset modules.

REQUIRED HARDWARE AND SOFTWARE

The following requirements are based on the simplest configurations with the best performance.

Hardware

CPU	450 MHz
RAM	256
Storage Space	20MB
Minimum Video	800 × 600

Software

Windows XP Professional
Internet Explorer Version 7.0

GLOSSARY

ID	Identification number (provided from the dropdown list)
Mb	Megabytes
MHz	Megahertz
WO	Work Order number (you choose your own for your records)

PARK INVENTORY

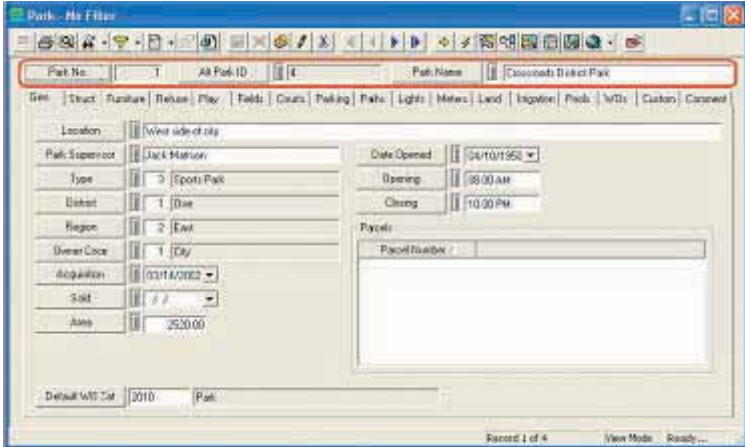
The Parks module is the most important module in the Parks Master. All other park asset modules are dependent upon the Parks module. While you can access the other asset modules through the gbaMS main menu, it's best to add park assets directly from the Parks module. This allows the system to carry over linking information and details about the park.

The following step-by-step instructions show you how to create a park record and report park inspections.

1. Open Trees/Parks>>Park Master>>Parks from the gbaMS main menu.
2. Click GO to bypass the filter.
3. When the module is opened, click the Add button  in the toolbar to access a blank record.
4. In the module header, create a unique Park Name and Alternate Park ID.

Note: Make sure to choose a name and ID that are recognizable. You'll need to be able to select the appropriate park name from a pick list in the park asset modules.

FIGURE 1 Name and ID



The screenshot shows the 'Park Master' form with the 'General' tab selected. The 'Park Name' field is highlighted with a red box. The form includes various fields for park information, including location, supervisor, type, district, region, owner code, acquisition date, cost, area, and opening/closing times. The 'Park Name' field contains the text 'Crossroads District Park'.

Note: In our example above, we've created a record for the Crossroads District Park and given it the Alternate Park ID, 4.

5. Record the park location, park supervisor, park type, district, region, owner, acquisition date, and opening date on the general tab.
6. Select a Default WO Category.

Note: The category you select here will be carried over to any work order created from this record.

INSPECTING PARK ASSETS

All of our park asset inventory modules have associated inspection modules. The following instructions show you how to add an inspection record for your park's swing set. This is an important consideration for liability reasons.

1. On the Inspections/Associations tab of the Park Playground Equipment Inventory module, right click in the Inspections grid and select Add Record.

FIGURE 2 Add Record

Equipment No. 1 All Equipment ID 14 Equipment Name Crossroads Swing Set

Default W/D Cat 2110 Park Playground Equipment Type 1 Swing

Attributes Inspections/Associations W/Ds Custom Comments

Inspections

Inspection Date	Overall Cond Test	Inspection Crew
Add Record		

Associations

Module Test	Asset No.	Alt ID
-------------	-----------	--------

Record 1 of 1 View Mode Ready...

Note: The Park Playground Equipment Inspection module will open. Here, you see the Equipment Number, Alternate Equipment ID, and Equipment Name carried over from the related inventory record.

FIGURE 3 Equipment Name and Number

Equipment No. 1 All Equipment ID 14 Equipment Name Crossroads Swing Set

Inspection Date 05/01/2007 Inspection Crew Oscar Next Inspection 05/01/2008

Condition 1 Condition 2 Custom Comments

Overall Cond	2	Good	Pinch Points	2	Good
Stability	1	Excellent	Fittings	2	Good
Concrete Exposure	2	Good	Fastenings	2	Fair
Protective Surface	2	Good	Nuts/Bolts	2	Good
Clearance	2	Good	Walls/Bedways	2	Good
Bar Support	2	Poor	Entrapments	1	Excellent
Leg Support	2	Good	Edges/Points	2	Good

Press Alt + Down Arrow for Calendar Record 1 of 1 Edit Mode Ready...

2. Enter the inspection date, inspection crew, and next scheduled inspection date.
3. Two Condition tabs are provided. Record the condition codes (excellent, good, fair, and poor) for the various aspects of the playground equipment. These include the overall condition, stability, concrete exposure, protective surface, clearance, supports, and fastenings, among other things.

FIGURE 4 Equipment Condition

Congratulations. Once you've saved the record, the Inspections grid in the Inventory module will be updated.

5

Source: Courtesy of gbaMS

CHAPTER HIGHLIGHTS

1. To organize your instruction effectively, include an introduction, a table of contents, and a discussion of sequenced steps.
2. Number your steps, and start each step with a verb.
3. Hazard alerts are important to protect your reader and your company. Place these alerts early in your instruction or before the appropriate step.
4. In the instruction, follow a chronological sequence.
5. Screencasting is a mixture of animated screen visuals, voiceovers, and captioned text. Screencasting benefits your audience by allowing the audience to see the procedure in action.
6. An SOP is a set of written instructions that documents routine or repetitive technical or administrative activities followed by business and industry. SOPs ensure accurate job performance and consistent quality and integrity of the end product. SOPs also ensure that governmental regulations are followed.
7. Highlighting techniques emphasize important points, thus minimizing damage to equipment or injury to users.
8. SOPs are used in many fields, including science, healthcare, biomedical technology, government, military, the computer industry, and others.
9. The writing process, complete with usability testing, will help you construct an effective instruction, user manual, or SOP.
10. Consider usability testing to determine if your procedure is valid.

APPLY YOUR KNOWLEDGE



CASE STUDIES

1. PhlebotomyDR provides instructions for its staff (nurses, doctors, and technicians) regarding the correct procedures to ensure cleanliness and employee protection. These include hand washing procedures and the correct use of sterile equipment, such as gloves, masks, aprons, and shoe coverings.

Following is a rough draft of one set of instructions for personnel safety. Revise the rough draft according to the criteria provided in this chapter. Correct the order of information, the grammar, and the instruction's content. In addition, improve the instruction by including appropriate graphics.

Hand washing

- Lather hands to cover all surfaces of hands and wrists.
- Wet hands with water.
- Rub hands together to cover all surfaces of hands and fingers. Pay special attention to areas around nails and fingers. Lather for at least 15 seconds.
- Dry thoroughly.
- Rinse well with running warm water.
- Avoid using hot water. Repeated exposure to hot water can lead to dermatitis.
- Use paper towels to turn off faucet.

Gloves

- Replace damaged gloves as soon as patient safety permits
- Don gloves immediately prior to task.
- Remove and discard gloves after each use.

Masks

Wear masks and eye protection devices (goggles or eye shields) to avoid droplets, spray, or splashes and to prevent exposure to mucous substances. Masks are also worn to protect nurses, doctors, and technicians from infectious elements during close contact with patients.

Aprons and Other Protective Clothing

Wear aprons or gowns to avoid contact with body substances during patient care procedures.

- Remove and discard aprons and other protective clothing before leaving work area.
- Some work areas might require additional protective clothing such as surgical caps and shoe covers or boots.

2. BurgerNet Online Order System is a software product in development. As a technical communicator, write the instruction to accompany this online order system.

BurgerNet Online Order System

1. **Description.** The BurgerNet application will be used to order burgers for delivery to a customer's home or work.

2. **Dropdown Menu Bar.** The dropdown menu bar includes a Task, Edit, View, and Help menu.
 - Task menu. On the “Task” menu customers will find a “Send Order” command, used to electronically send the completed order to the restaurant. The “Print Order” command on the “Task” menu lets the customer print a copy of the order. The “Exit” command terminates the order session.
 - Edit menu. On the “Edit” menu, an “Add Favorites” command allows customers to add a frequently ordered combination of food items to make future online ordering quick and easy. The “Edit Favorites” command lets customers modify an existing favorite order.
 - View menu. On the “View” menu, the “Preferences” command is used to modify default customer. In addition, when favorites have been defined, they are listed under “View.” Customers can select a favorite for display before sending the order.
 - Help menu. On the “Help” menu, customers can access FAQs.
3. **Key Components of BurgerNet.**
 - Customer Info. frame: Customers type in their name and telephone number in the “Name” and “Extension” fields.
 - Burgers frame. In the “Burgers” fields customers specify the number of burgers desired. The “Cheese” dropdown list lets the customer select either American cheese, Pepperjack, Mozzarella, or Swiss cheese to top their burgers. “Bun” lets customers choose either plain or onion buns from the dropdown list.
 - Toppings. Here, customers check what to put on their burgers.
 - Sides: Select which side dish or dishes and specify the quantity for each.
 - Beverages: Select drink(s) and specify the quantity for each.
 - Total field. The total cost of the customer's order is automatically displayed.

INDIVIDUAL AND TEAM PROJECTS

1. Write an instruction or SOP. To do so, first select a topic. You can write an instruction telling how to monitor, repair, test, package, plant, clean, operate, manage, open, shut, set up, maintain, troubleshoot, install, use software, and so on. Choose a topic from your field of expertise or one that interests you. Follow the writing process techniques to complete your instruction. Prewrite (using a flowchart), write a draft (abiding by the criteria for instructions or SOPs presented in this chapter), and rewrite to perfect your text.
2. Find examples of instructions, user manuals, or SOPs written in the work environment. Bring these to class. Using the criteria for instructions and SOPs presented in this chapter, decide whether the instructions and SOPs are successful or unsuccessful. If the instructions and SOPs are good, show how and why. If they are flawed, explain the problem(s). Then rewrite the instructions or SOPs to improve them.
3. Good writing demands revision. Following is a flawed instruction. To improve it, rewrite the text, abiding by the criteria for instructions and the rewriting techniques included in this chapter.

Date: November 1, 2014
To: Maintenance Technicians
From: Second Shift Supervisor
Subject: Oven Cleaning

The convection ovens in kiln room 33 need extensive cleaning. This would consist of vacuuming and wiping all walls, doors, roofs, and floors. All vents and dampers need to be removed and a tack cloth used to remove loose dust and dirt. Also, all filters need replacing. I am requesting this because when wet parts are placed in the ovens to cure the paint, loose particles of dust and dirt are blown onto the parts, which causes extensive rework. I would like this done twice a week to ensure cleanliness of product.

PROBLEM-SOLVING THINK PIECE

Read the following instructional steps. Are they in the correct chronological order? How would you reorder these steps to make the instruction more effective?

Changing Oil in Your Car

Run the car's engine for approximately 10 minutes and then drain the old oil.
Park the car on a level surface, set the parking brake, and turn off the car's engine.
Gather all of the necessary tools and materials you might need.
Open the hood.
Jack up and support the car securely.
Place the funnel in the opening and pour in the new oil.
Replace the cap when you have finished pouring in new oil.
Locate the oil cap on top of the engine and remove the cap.
Tighten the plug if you find leakage.
Run the engine for a minute, then check the dipstick. Add more oil if necessary.
Pour the used oil into a plastic container and dispose of it safely and legally.

WEB WORKSHOP

- Review any of the following Web sites' online instructions. Based on the criteria provided in this chapter, are the instructions successful or not?
 - If the answer is yes, explain why and how the instructions succeed.
 - If the answer is no, explain why the instructions fail.
 - Rewrite any of the flawed instructions to improve them.

Web Sites	Topics
http://www.hometips.com/diy.html	Electrical systems, plumbing, kitchen appliances, walls, windows, roofing, and siding
http://www.hammerzone.com	Kitchen projects, tubs, sinks, toilets, showers, and water heaters
http://dmoz.org/Home/Home_Improvement/	Links to step-by-step procedures for painting, welding, soldering, plumbing, building walls, installing windows, and repairing and installing doors
http://directory.google.com/Top/Home/Home_Improvement/	Links to sites for instructions on decorating, electrical work, flooring installation, furniture selection, lighting, painting, plumbing, welding, window installation, and door repair
http://www.quakerstate.com/#/car-care	Instructions for car care

- The U.S. Department of Labor article "Hazard Communication: A Review of the Science Underpinning the Art of Communication for Health and Safety" can be found at <http://www.osha.gov/dsg/hazcom/hc2inf2.html>. This article focuses on many important aspects of writing instructions, including the use of icons, readability, and audience variables. Access the article and report your findings either orally or in a memo, letter, or e-mail message.
- YouTube provides thousands of instructional videos for topics including creating Origami shapes, playing World of Warcraft, playing pool and billiards, snowboarding and paragliding, playing guitars, tossing pizzas, using Macs and PCs, installing software and hardware, and downloading music for iPods. Research any of these videos (or others). Based on the criteria provided in this chapter, assess your findings. Were the videos successful? How could they be improved? Share your findings with your teacher in an e-mail message or present your findings to your class in a brief oral presentation.

TEXT CREDITS

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Table 1 — Text without Graphics for High-Tech readers.
Courtesy of Nelcor Puritan Bennett.

Table 2 — Text without Graphics for High-Tech readers.
Courtesy of Covidien, LLC.

Figure 4 — Instructions with Large, Bold Graphics for
Low-Tech Audience. Courtesy of Puritan Bennett Corp.

Figure 4 — Instructions with Large, Bold Graphics for
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Figure 7 — HP Officejet Pro K550 Series Color Printer. ©
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Figure 10 — gbaMS Instruction Rough Draft with Peer
Reviewer Commentary. Courtesy of George Butler
Associates.

Figure 11 — gbaMS Parks Inventory Completed
Instruction. Courtesy of gbaMS.

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REFERENCES

Sharp, Roger A. "Incorporating Animation into Help
Files." *Intercom* Nov. 2007: 8–12. Print.

Tietjen, Phil. "Adventures in Screencasting." *Intercom* Jul./
Aug. 2008: 8–10. Print.